McMaster University ENGINEERING | mohawk | W BOOTH SCHOOL OF ENGINEERING PRACTICE AND TECHNOLOGY
McMaster-Mohawk Bachelor of Technology Partnership

Laboratory 4
Load Cells

SMRTTECH 3CC3

Cloud Computing and Internet of Things

B.Tech - PROCTECH

PRELAB

Please read the following material prior to this lab:

1. Read the datasheets of the LMC6482 CMOS Quad Operational Amplifier and the load cell. Both can be downloaded from Avenue to Learn.

OBJECTIVES

In this laboratory, you will build an **analog circuit** that will enable you to use a **load cell** to measure the **force** of a metal bar. You will then use **operational amplifiers** to amplify the voltage output from your circuit.

MATERIALS

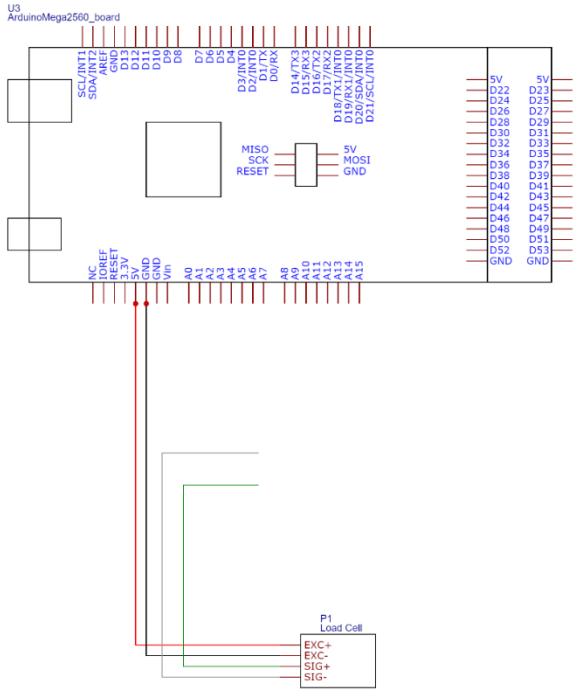
1. Arduino Mega
2. Breadboard
3. Jumper wires
4. 2x 100k Ohms resistors
5. 2x 1M Ohms resistors
6. 2x 10k Ohms resistors
7. Load Cell & Screw Terminals
8. LMC6484 Op-Amp

PROCEDURE

Exercise 1: Build an Analog Circuit for Obtaining Force

In this exercise, you will observe the voltage output from your load cell circuit using a voltmeter.

1. Build the analog circuit shown in the **figure 1**. Since we will be continually adding to the circuit it is best if the initial circuit is constructed at the far end of the breadboard.
    a. In all the following exercises, use the **Arduino 5V pin** for as the source voltage, and the **GND** pin as the ground.
    b. You may have to use a screwdriver to screw in your wires to the **screw terminals**.
    c. Make sure to **plug in** your **USB** into the **Arduino** to power the circuit.



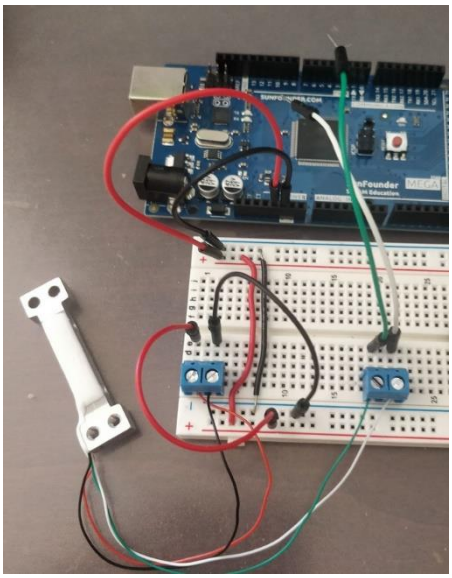


Figure 1. Analog sensor circuit for using a load cell as a Force Sensor

2. Using a voltmeter, measure the drop across SIG+ and SIG-. Put the load cell at the edge of your table and pin one end of the cell with your hand. Carefully apply force to the other end of the load cell with your other hand. Make sure to apply the force in the same direction as shown on the side of the cell.



3. Record the voltage range of the load cell (when **no pressure** is applied and when **maximum pressure** is applied).

End of Exercise

Exercise 2: Add a first stage amplifier

In this exercise, you will add the first stage amplifier to your circuit.

1. The voltage output from the sensor will be amplified by the LMC6484 operational amplifier.

2. The LMC6484 IC has a total of 14 pins. In this exercise you will use **pin 2** for the SIG- (-) signal, **pin 3** for the SIG+ (+) signal, and **pin 1** for output.

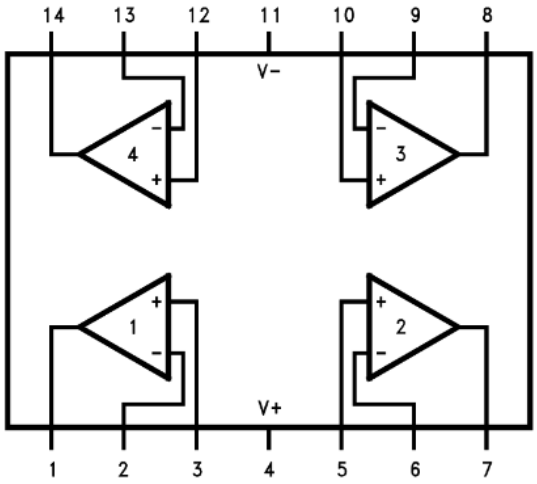


Figure 2. LMC6484 IC Pin layout

3. Connect the Op-Amp, Arduino, and sensor as shown in the **Figure 3**.
    a. Choose the op-amp resistors such that $R_3 = R_4 = 1$ MΩ, and $R_1 = R_2 = 10$ kΩ.
    b. The voltage gain *AV* of your op-amp circuit will be **AV = R3 / R1 = 100.**
    c. Connect V2 to **Analog pin 0**.

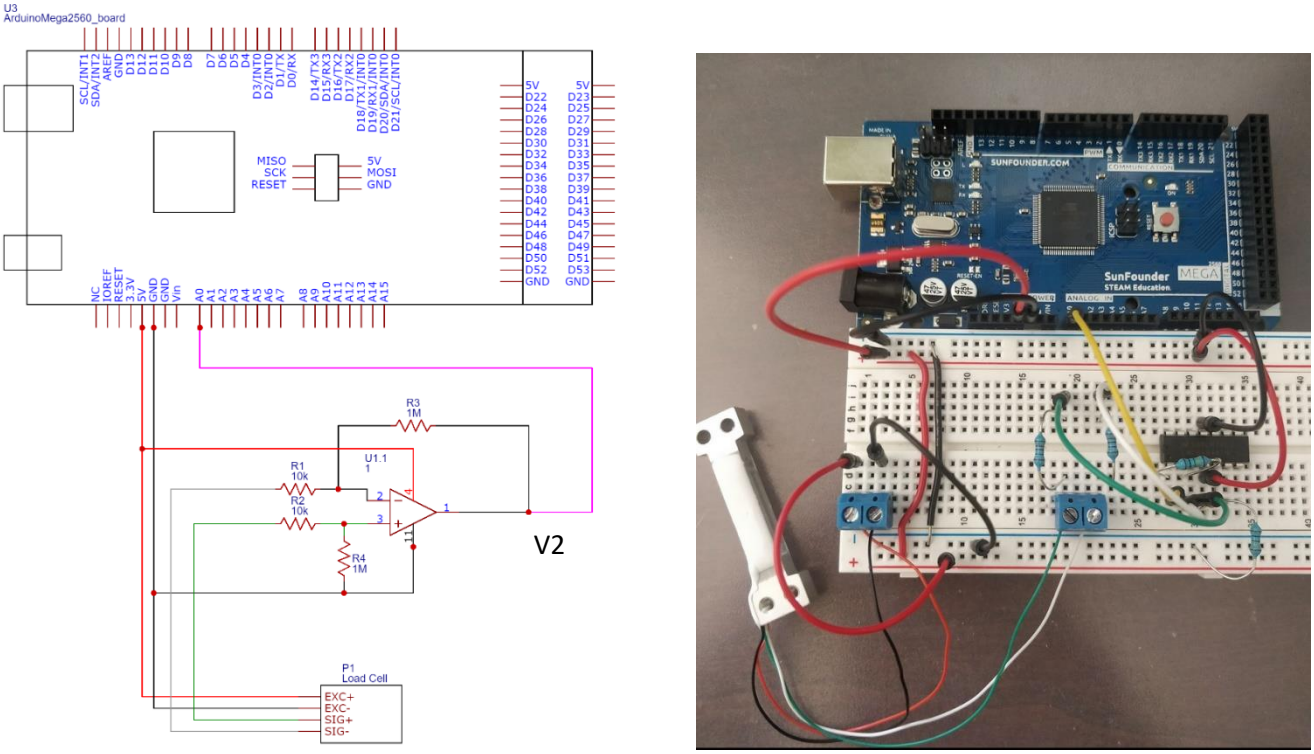


Figure 3. First Stage Op-Amp Circuit

4. Repeat steps 2 & 3 from the previous exercise, except this time, measure the voltage drop across V2 (from **Figure 3**) and **GND**. Record the range of amplified voltage.

End of Exercise

Exercise 3: Add a second stage amplifier

In this experiment, you will amplify the voltage output from your circuit using an additional non-inverting operational amplifier.

This amplification is necessary to increase the percentage accuracy of the voltage measurements made by the analog pin of the Arduino. One way to amplify the voltage output is to **increase** the resistance of both R_3 and R_4 or to **decrease** the resistance of both R_1 and R_2. We will assume that the circuit you built in **Exercise 2** cannot be modified, which is usually the case with many circuit boards you may encounter in industrial applications. Instead, amplification of the existing voltage output **V2** could be done quite easily using a non-inverting op-amp.

1. Add the circuit shown in **Figure 4** to the circuit you built in Exercise 2.
    a. Use op-amp number three (pin numbers 8, 9 and 10) on the LMC6484 chip (Figure 2).
    b. Choose the source resistor R_5 and the feedback resistor R_6 such that R_5 = 10 kΩ R_6 = 100 kΩ.
    c. The additional voltage gain *AV* of your op-amp circuit will be **AV = 1 + (R_6 / R_5) = 11.** The amplified voltage should be 11 times greater than what was viewed in Exercise 2.

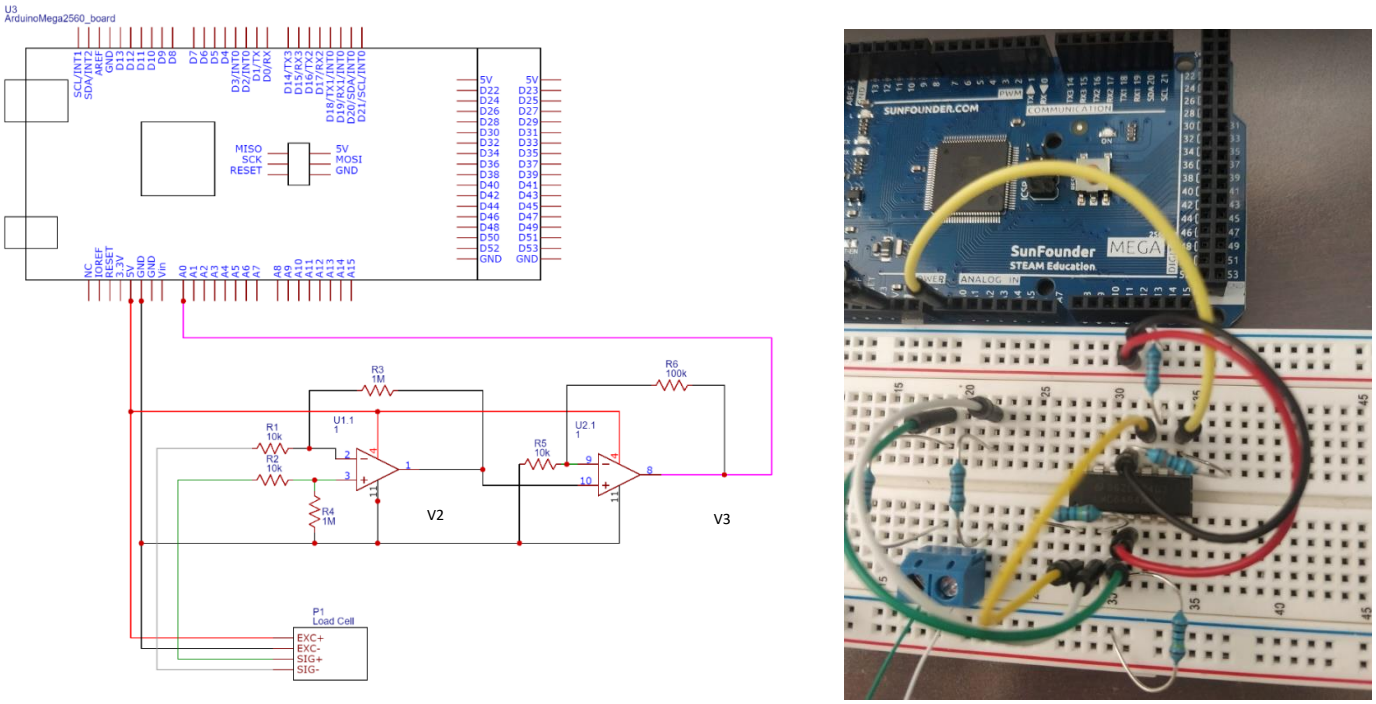


Figure 4. Second Stage Op-Amp Circuit

2. Repeat steps 2 & 3 from the Exercise 1, except this time, measure the voltage drop across V3 (from **Figure 4**) and **GND**. Record the range of amplified voltage.

End of Exercise

Exercise 4: View Load Cell Output on Computer Screen

In this experiment, you will build a LabVIEW VI to measure and display the voltage output from the strain gauge sensor circuit you built in Experiment #2.

1. Create a new project and open the **Front Panel** window.

2. Add a **Numeric Indicator** control from **Controls»Modern»Numeric.** Name the control `Voltage Reading`.

3. Switch to the **block diagram** window.

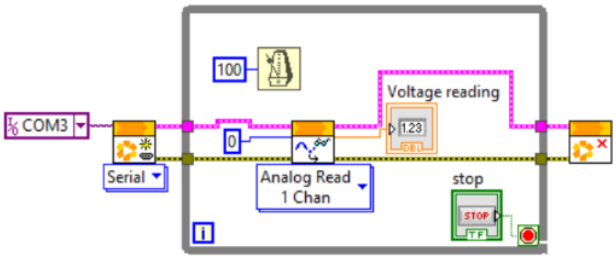
4. Add the **LINX Open** function from the **Functions»MakerHub»LINX** palette.
   a. Right click on the `Serial Port` node of the **LINX Open** function and select **Create constant.**
   b. Select your comm port from the drop down.

5. Add a **LINX Analog Read** function from the **Functions»MakerHub»LINX»Peripherals»Analog** palette.
   a. Right click on the `Analog Channel` node of the **LINX Analog Read** function and select **Create constant.**
   b. Write `0` into the constant.

6. Add the **LINX Close** function from the **Functions»MakerHub»LINX** palette.

7. Add a **While Loop** from **Functions»Programming»Structure.** Encompass the loop around everything except for the **LINX Open & Close** functions.
   a. Add a **stop** button to the **While Loop**.
   b. Add a **Wait Until Next ms Multiple**  inside the **While Loop** from **Functions»Programming»Timing.**
   c. Set the `millisecond multiple` node of the **Wait Until Next ms Multiple** function to a numeric constant `100`.

8. Wire the functions as shown in **Figure 5**.



9. **Figure 5**. LabVIEW VI

10. Run the program. Make sure to use the **Stop button** to stop the program.

11. Repeat Step 2 from Exercise 3. You should see the same voltage range.

End of Exercise

Extend your knowledge

1. What is the total voltage amplification of your circuit? Modify the circuit such that the amplification if **halved**. Draw a schematic of the new circuit. Measure the new voltage range.

Laboratory Report

the lab report must follow the **"REPORT WRITING For IoT.pdf"** document which is posted on avenue to learn. It should also include the following

1. All the values that were measured during the lab procedure in a tabulated form.
2. Explain what a Wheatstone Bridge is. How is it used in load cells?
3. State the role of the Op-amps in your circuit.
4. Explain how a pressure transducer uses strain gauges